

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 1

For the tutorial in Week 2 on 25 January, let us discuss

- Ex. 6.1.3, 6.1.4, 6.1.6, 6.1.7, 6.1.9, 6.1.10, 6.1.13, 6.1.13, 6.1.20, 6.1.21.

Ex. 6.1.3 and Ex. 6.1.14.

- (i) Assume there are 26 letters, how many different three-letter initials can people have ?
- (ii) A committee is formed consisting of one representative from each of the 50 states in the United States, where the representative from a state is either the governor or one of the two senators from that state, How many ways are there to form this committee ?

Solution:

- (i) $26^3 = 17576$.
Each letter has 26 choices, hence by product rule 26^3 for three-letters.
- (ii) 3^{50} .
For each state, there are 3 choices (by sum rule), and there are 50 states, by product rule 3^{50} .

Ex. 6.1.6 and 6.1.7.

- (i) How many bit strings with length not exceeding n , where n is a positive integer, consist entirely of 1's, not counting the empty string ?
- (ii) How many bit strings of length n , where n is a positive integer, start and end with 1's ?

Solution:

- (i) n .
For any fixed m with $1 \leq m \leq n$, there is exactly one string consisting entirely of 1's, that is the all '1' string of length m , and there are n such m 's fulfilling $1 \leq m \leq n$.
- (ii) 2^{n-2} for $n \geq 2$, and 1 for $n = 1$.
The start and end bit each has 1 choice for the bit, which is "1", for the rest $n - 2$ bits, each has 2 choices (i.e., can be either "0" or "1"), hence by product rule $1 \cdot 2 \cdot 2 \cdots 2 \cdot 1 = 2^{n-2}$.

Ex. 6.1.10. There are 4 choices $\{G, U, A, C\}$ for each element of an RNA sequence, how many 6-element RNA sequences

- (a) do not contain U ?
- (b) end with GU ?
- (c) start with C ?
- (d) contain only A or U ?

Solution:

- (a) 3^6 .
Element has $4 - 1 = 3$ choices, there are 6 elements, by product rule the answer is 3^6 .

(b) 4^4 .

Each of the first 4 elements has 4 choices, and each of the last 2 elements has 1 choice (i.e., the second last element has the only choice to be “G” and the last element has the only choice to be “U”). By product rule, the answer is $4 \cdot 4 \cdot 4 \cdot 4 \cdot 1 \cdot 1 = 4^4$.

(c) 4^5 .

Each of the last 5 elements has 4 choices, and the first element has the only one choice to be “C”. By product rule, the answer is $1 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 = 4^5$.

(d) 2^6 .

Element has 2 choices (either “A” or “U”), there are 6 elements, by product rule the answer is 2^6 .

Ex. 6.1.12 (a)–(c). How many positive integers between 100 and 999 (both inclusive)

(a) are divisible by 7 ?

(b) are odd ?

(c) have the same three decimal digits ?

Solution:

$$(a) \left\lfloor \frac{999}{7} \right\rfloor - \left\lfloor \frac{100}{7} \right\rfloor = 142 - 14 = 128.$$

$$(b) \left\lceil \frac{999}{2} \right\rceil - \left\lceil \frac{100}{2} \right\rceil = 500 - 50 = 450.$$

(c) 9.

Ex. 6.1.12 (d)–(h). How many positive integers between 100 and 999 (both inclusive)

(d) are not divisible by 4 ?

(e) are divisible by 3 or 4 ?

(f) are not divisible by either 3 or 4 ?

(g) are divisible by 3 and 4 ?

(h) are divisible by 3 but not by 4 ?

Solution:

For any positive integer n , let A_n be the set of positive integers between 100 and 999 (both inclusive) that are divisible by n . For a set S , we use S^c to denote its complement set. Then

$$|A_3| = \left\lfloor \frac{999}{3} \right\rfloor - \left\lfloor \frac{99}{3} \right\rfloor = 333 - 33 = 300,$$

$$|A_4| = \left\lfloor \frac{999}{4} \right\rfloor - \left\lfloor \frac{99}{4} \right\rfloor = 249 - 24 = 225,$$

$$|A_3 \cap A_4| = |A_{12}| = \left\lfloor \frac{999}{12} \right\rfloor - \left\lfloor \frac{99}{12} \right\rfloor = 83 - 8 = 75.$$

$$(d) |A_4^c| = (999 - 99) - |A_4| = 900 - 225 = 675.$$

$$(e) |A_3 \cup A_4| = |A_3| + |A_4| - |A_3 \cap A_4| = 300 + 225 - 75 = 450.$$

$$(f) |(A_3 \cup A_4)^c| = (999 - 99) - |A_3 \cup A_4| = 900 - 450 = 450.$$

$$(g) |A_3 \cap A_4| = |A_{12}| = 75.$$

$$(h) |A_3 \cap A_4^c| = |A_3| - |A_3 \cap A_4| = 300 - 75 = 225.$$

Ex. 6.1.13. How many strings of three decimal digits

- (a) do not contain the same digit three times ?
- (b) begin with an odd digit ?
- (c) have exactly two digits that are 4's ?

Solution:

- (a) $10^3 - 10 = 990$. There are 10 choices for bit position, and there are 3 positions, hence in total $10^3 = 1000$ such strings. Among them, there are 10 with the same digit three times.
- (b) $5 \cdot 10 \cdot 10 = 500$. There are 5 choices for the first position, and 10 for each of the last two positions.
- (c) $3 \cdot 9 = 27$. There is one digit not to be 4, there are 3 choices to place this digit, and this digit has 9 choices, the rest two digits have exactly one choice (i.e., 4). By product rule, the answer is $3 \cdot 9$.

Ex. 6.1.9. There are 128 different ASCII characters, how many strings of 5 ASCII characters contain the character '@' at least once ?

Solution: There are in total 128^5 length-5 strings without any restriction, and 127^5 length-5 strings containing no '@', hence the answer is $128^5 - 127^5 = 1321368961$. $\square$

Ex. 6.1.20 and 6.1.21. Consider functions that maps from the set $A = \{1, 2, \dots, n\}$ to the set $B = \{0, 1\}$, where n is a positive integer,

- (a) how many such functions are there ?
- (b) how many such functions are one-to-one?
- (c) how many such functions are there that assign 0 to both 1 and n ?
- (d) how many such functions are there that assign 1 to exactly one element in A ?

Solution:

- (a) 2^n .
Each element from A has 2 choices (0 or 1) under the mapping, and there are n elements in A, by product rule the answer is 2^n .
- (b) If $n = 1$, then 1 can be either mapped to 0 or 1. So there are two such functions.

If $n = 2$, then again 1 can be either mapped to 0 or 1, and then the choice of 2 will be fixed. Thus we have two such functions.

If $n = 3$, then it is not possible to construct any one-to-one function.

- (c) 2^{n-2} .
For 1 and n from A, there is only one choice under the mapping, for the rest $n - 2$ elements in A, there are 2 choices for each.
- (d) There are n choices to select the element from A to be assigned to 1, and the rest elements from A has the only choice ("0") left under the assignment. Hence n choices in total.